Cognitive Modeling

Complete Lecture Notes

Week 4: Maximum Likelihood Estimation

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Chapter 1

Parameter Estimation Methods

1.1 Week 4: Maximum Likelihood Estimation

Chapter 2: Parameter estimation methods · Lecture 1 h · Lab 2 h Running data: the Week 2–3 forgetting experiment (binomial recall counts), plus a new SDT experiment with three payoff conditions.

1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

English	فارسی	توضیح کوتاه
likelihood	درست‌نمایی	احتمال داده به‌عنوان تابعی از پارامترها
estimate / estimator	برآورد / برآوردگر	روش به / مقدار به‌دست‌آمده برای پارامتر دست آوردن آن
parameter	پارامتر	کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند
forgetting curve	منحنی فراموشی	کاهش یادآوری با گذشت زمان
binomial distribution	توزیع دوجمله‌ای	توزیع تعداد موفقیت در n کوشش
signal detection theory (SDT)	نظریه‌ی تشخیص علامت	مدلی که حساسیت را از سوگیری پاسخ جدا می‌کند
log-likelihood / NLL	لگاریتم درست‌نمایی / منفی آن	مقداری که بهینه‌سازها کمینه یا بیشینه می‌کنند
least squares	کمترین مربعات	روش برازش با کمینه کردن مجموع مربع خطاها

English	فارسی	توضیح کوتاه
maximum likelihood estimate (MLE)	برآورد بیشینه‌ی درست‌نمایی	پارامترهایی که داده را محتمل‌ترین می‌کنند
noise	نویز	تغییرات تصادفی در شواهد یا در داده
fit / fitting	برازش	تنظیم پارامترها تا مدل با داده جور شود
deviance (G^2)	انحراف (deviance)	معیار عدم برازش نسبت به مدل اشباع‌شده
bounds / constraints	قیدها / کران‌ها	محدودیت‌های مجاز برای مقدار پارامترها
response bias	سوگیری پاسخ	تمایل به یک پاسخ، مستقل از محرک
sensitivity (d′)	حساسیت	فاصله‌ی توزیع علامت و نویز
trial	کوشش (trial)	یک بار انجام تکلیف توسط شرکت‌کننده
optimizer	بهینه‌ساز	الگوریتمی که کمینه‌ی تابع را پیدا می‌کند
SSE / RMSD	ریشه‌ی میانگین / مجموع مربعات خطا مربع انحراف	معیارهای فاصله‌ی داده و پیش‌بینی
saturated model	مدل اشباع‌شده	مدلی که داده را دقیقاً بازتولید می‌کند
likelihood-ratio test (LRT)	آزمون نسبت درست‌نمایی	مقایسه‌ی دو مدل تودرتو
exponential / power function	توانی / تابع نمایی	دو شکل رایج برای منحنی فراموشی و یادگیری
liberal / conservative	محافظه‌کار / آسان‌گیر	«نه» / تمایل به گفتن «بله»
token	token (واحد متن)	کلمه یا بخشی از کلمه که مدل زبانی پیش‌بینی می‌کند
perplexity	perplexity (سردرگمی)	e به توان هر token NLL

1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Distinguish probability (data varying, parameters fixed) from likelihood (data fixed, parameters varying).
2. Derive the likelihood and log-likelihood for binomial (Bernoulli) and Gaussian data models, and show that least squares is maximum likelihood under equal-variance Gaussian noise.
3. Implement maximum likelihood estimation (MLE) by minimizing the negative log-likelihood (NLL) in Python, for forgetting curves and for a multi-condition SDT model.
4. Evaluate absolute goodness of fit with the deviance G^2 and compare MLE with least-squares estimates.
5. Interpret constrained models (e.g., “payoffs change bias, not sensitivity”) as substantive psychological hypotheses.

1.1.3 Motivation & big picture

In Week 3, least squares treated every data point as equally reliable. But a recall proportion of 0.50 measured on 50 items is *noisier* than a proportion of 0.95: binomial variance $f(1-f)/n$ depends on f. Least squares also has nothing to say about data that are not continuous numbers, such as a single yes/no response, a choice among options, or a confidence rating.

Maximum likelihood replaces "make predictions close to data" with a sharper question:

> Which parameter values make the data we actually observed most probable?

To answer it, the model must say how probable *any* possible dataset is, which is exactly the noise model (component 4 of Week 2's anatomy). Once we have it, MLE handles counts, choices, RTs and mixtures of these in the same way. It also gives us, for free, an absolute measure of misfit (the deviance), the basis for standard errors (Week 5), information criteria (Week 7), and the likelihood term in Bayes' rule (Weeks 9–13). The likelihood is the single most important object in this course.

Where is this used?

- Almost every published cognitive model is fitted by maximum likelihood or by its Bayesian extension (Weeks 9–13). When a paper reports "model fits", "AIC" or "log-likelihood", MLE is underneath.
- Choice data: which option a person picks (in a memory test, a gamble, a learning task) is not a continuous number, so least squares does not apply. The likelihood handles choices naturally.
- Psychophysics: the psychometric function (probability of detecting a stimulus as a function of its strength) is fitted by binomial MLE in every vision and hearing lab.
- Testing hypotheses inside a model: "does the payoff change sensitivity or only bias?" becomes a comparison of two likelihoods (Section 6).
- AI: large language models are trained by maximizing the likelihood of text (see the AI & LLMs section).

1.1.4 Lecture notes

In plain words Imagine trying different parameter values one by one. For each value, ask the model: "if this were the true value, how probable would the data I actually collected be?" That number is the likelihood of that parameter value. The maximum likelihood estimate (MLE) is the parameter value under which the observed data would have been most probable. Example: someone recalls 7 of 10 words. If their true recall probability were 0.5, getting exactly 7 would be fairly unlikely (0.117); if it were 0.7, it would be more likely (0.267). So 0.7 is the better explanation, and indeed the MLE is 7/10. In practice we work with the logarithm of the likelihood (so products become sums) and we let the computer *minimize* its negative (NLL), using the optimizers of Week 3. Two bonuses: least squares turns out to be a special case of MLE, and the likelihood gives an absolute test of misfit (G^2).

Suggested timing (60 min): probability vs. likelihood 10 · binomial MLE 15 · log-likelihood and NLL 10 · Gaussian case (LS = MLE) 10 · goodness of fit G^2 10 · constrained models 5.

1. Probability vs. likelihood

The same function $p(\mathrm{y} \mid \boldsymbol{\theta})$ can be read in two ways:

Reading	Fixed	Varies	Name	Sums/integrates to 1?
Before the experiment	$\boldsymbol{\theta}$	y	Probability (mass/density) of data	Yes, over y
After the experiment	y (observed)	$\boldsymbol{\theta}$	Likelihood $L(\boldsymbol{\theta}) = p(\mathrm{y} \mid \boldsymbol{\theta}, \mathcal{M})$	No, not over $\boldsymbol{\theta}$

The likelihood is not the probability that the parameters are correct. It is the probability of the data *under* each candidate parameter value. (Turning it into a probability distribution over $\boldsymbol{\theta}$ requires a prior: Bayes, Week 9.)

Example. A participant recalls $y = 7$ of $n = 10$ items. Under a binomial model with recall probability θ:

$$L(\theta) = \binom{10}{7} \theta^7 (1-\theta)^3.$$

$L(0.5) = 0.117$, $L(0.7) = 0.267$: the observed data are about 2.3 times more probable if $\theta = 0.7$ than if $\theta = 0.5$.

Reading the math $L(\theta) = \binom{10}{7}\theta^7(1-\theta)^3$:

- θ^7: the probability that 7 specific items are recalled (each with probability θ).
- $(1-\theta)^3$: the probability that the other 3 are forgotten.
- $\binom{10}{7} = 120$ ("10 choose 7"): the number of different ways to pick which 7 of the 10 items are the recalled ones. It does not depend on θ, so it never changes *where* the maximum is.
- Plug in $\theta = 0.7$: $120 \times 0.7^7 \times 0.3^3 = 120 \times 0.0824 \times 0.027 = 0.267$.

The key mental switch: in probability, θ is fixed and we ask about different data; in likelihood, the data are fixed (we already have them) and we compare different θ.

2. The maximum likelihood estimate

$$\hat{\boldsymbol{\theta}}_{\mathrm{ML}} = \arg\max_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}), \qquad \ell(\boldsymbol{\theta}) = \ln L(\boldsymbol{\theta}).$$

Why the log? 1. With independent observations the likelihood is a product, $L = \prod_i p(y_i \mid \boldsymbol{\theta})$, which underflows to 0 on a computer for even a few hundred trials. The log turns it into a sum, $\ell = \sum_i \ln p(y_i \mid \boldsymbol{\theta})$. 2. The log is strictly increasing, so it has the same maximizer. 3. Sums are easier to differentiate.

Optimizers *minimize*, so in code we minimize the negative log-likelihood, $\mathrm{NLL}(\boldsymbol{\theta}) = -\ell(\boldsymbol{\theta})$.

Analytic example (binomial). $\ell(\theta) = \text{const} + y \ln \theta + (n - y) \ln(1 - \theta)$. Setting the derivative to zero:

$$\frac{d\ell}{d\theta} = \frac{y}{\theta} - \frac{n - y}{1 - \theta} = 0 \quad \Rightarrow \quad \hat{\theta} = \frac{y}{n}.$$

The MLE of a probability is the observed proportion, which is reassuring. For cognitive models the predicted probability is a *function* of the parameters, $\theta_i = f(x_i; \boldsymbol{\theta})$, and we usually need a numerical optimizer.

3. The binomial likelihood for forgetting curves

With y_i items recalled out of n_i at interval t_i, and independent intervals:

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{N} \left[\ln \binom{n_i}{y_i} + y_i \ln f(t_i; \boldsymbol{\theta}) + (n_i - y_i) \ln \big(1 - f(t_i; \boldsymbol{\theta})\big) \right].$$

- The binomial coefficient does not depend on $\boldsymbol{\theta}$, so it does not affect the MLE (but keep it if you want to compare log-likelihoods across different data codings; `scipy.stats.binom.logpmf` includes it).
- Observations predicted to be near 0 or 1 have small variance, so a prediction that misses them is punished more heavily. MLE performs the variance weighting of Week 3's weighted least squares *automatically*.
- Numerical safety: if the model predicts $f = 0$ where $y > 0$, then $\ln 0 = -\infty$. Clip predictions to $[\epsilon, 1 - \epsilon]$ with, e.g., $\epsilon = 10^{-10}$.

Bernoulli vs. binomial. If we keep the individual 0/1 responses, $\ell = \sum_{\text{trials}} [y \ln f + (1 - y) \ln(1 - f)]$. This differs from the binomial version only by the constant binomial coefficients, so it gives the same MLE.

Reading the math The forgetting log-likelihood is the single-proportion log-likelihood from Section 2, written once per retention interval and added up:

- for interval i: $y_i \ln f(t_i)$ rewards predicting a high recall probability where many items were recalled; $(n_i - y_i) \ln(1 - f(t_i))$ rewards predicting a low recall probability where many were forgotten.
- Adding over intervals is allowed because the intervals are assumed independent: the log of a product of probabilities is the sum of the logs.
- Why it weights points automatically: if the model predicts $f = 0.95$ but only 40 of 50 are recalled, the term $10 \ln(0.05) \approx -30$ is very costly. Near $f = 0.5$, the same absolute miss costs much less. Points that *should* be precise are trusted more.

4. The Gaussian case: least squares is maximum likelihood

Suppose $y_i = f(x_i; \boldsymbol{\theta}) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma)$, independent and with the same σ. Then

$$\ell(\boldsymbol{\theta}, \sigma) = \sum_{i=1}^{N} \ln \left[\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - f(x_i; \boldsymbol{\theta}))^2}{2\sigma^2} \right) \right] = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{\mathrm{SSE}(\boldsymbol{\theta})}{2\sigma^2}.$$

For any fixed σ, maximizing ℓ over $\boldsymbol{\theta}$ means minimizing SSE. So least squares = MLE under i.i.d. equal-variance Gaussian noise. Maximizing over σ as well gives $\hat{\sigma}^2 = \mathrm{SSE}/N$.

This explains *when* least squares is appropriate (continuous, roughly symmetric noise with constant variance) and when it is not (counts, proportions, choices, skewed RTs).

> Advanced: properties of the MLE. Under regularity conditions (identifiable model, true parameters not on the boundary, N large), the MLE is *consistent* (converges to the true value), *asymptotically normal*, and *asymptotically efficient* (no consistent estimator has smaller variance in the limit). Its sampling variance is approximated by the inverse of the Fisher information, the curvature of $-\ell$ at its minimum. Week 5 uses exactly this to compute standard errors. In small samples the MLE can be biased (e.g., $\hat{\sigma}^2 = \mathrm{SSE}/N$ underestimates σ^2).

5. Absolute goodness of fit: the deviance G^2

The log-likelihood value alone has no absolute meaning. To judge it, compare the model with the saturated model, which has one free probability per data cell and therefore reproduces every observed proportion exactly ($\hat{f}_i = y_i/n_i$). The deviance is

$$G^2 = 2\big[\ell_{\text{saturated}} - \ell(\hat{\boldsymbol{\theta}})\big] = 2 \sum_i \left[y_i \ln \frac{y_i}{n_i \hat{f}_i} + (n_i - y_i) \ln \frac{n_i - y_i}{n_i(1 - \hat{f}_i)} \right],$$

with $0 \ln 0 = 0$. If the model is correct and counts are not too small, G^2 approximately follows a χ^2 distribution with df $=$ (number of cells) $- k$. A large G^2 (small p) means the model misfits *more than sampling noise can explain*.

Caveats: with very large N, any small misfit becomes "significant"; with small N, real misfit can go undetected. A non-significant G^2 does not prove the model is right (Week 6).

Reading the math

- The saturated model is the "cheating" model that predicts every observed proportion exactly. Its log-likelihood is the best any model could possibly achieve on these data.
- $G^2 = 2(\ell_{\text{saturated}} - \ell(\hat{\boldsymbol{\theta}}))$ measures how far our model falls short of that ceiling. $G^2 = 0$ would be a perfect fit.
- Degrees of freedom = number of data cells − number of fitted parameters. With 8 intervals and 2 parameters, df = 6.

- If the model is true, G^2 behaves like a χ^2 variable with that df, whose average is df. So $G^2 \approx 6$ with df = 6 is a normal amount of misfit (just noise); $G^2 = 26$ with df = 6 ($p = .0002$) means the model systematically misses the data.

6. Constrained models as hypotheses

MLE makes it easy to fit several conditions at once with some parameters shared. Example: an SDT experiment with three payoff conditions (rewarding "yes" responses, neutral, rewarding "no").

Model	Free parameters	k	Hypothesis
Shared sensitivity	$d'; c_1, c_2, c_3$	4	Payoffs change *bias* only
Separate	$d'_1, d'_2, d'_3; c_1, c_2, c_3$	6	Payoffs may also change *sensitivity* (this is the saturated model for 6 rates)

The joint log-likelihood is the sum over conditions of the binomial log-likelihoods of hits and false alarms. Comparing the two models' maximized log-likelihoods tests a substantive question. With nested models like these, $2(\ell_6 - \ell_4)$ is approximately χ^2 with 2 df: the likelihood-ratio test, developed in Week 7.

> Advanced: SDT MLE and the formulas. For a single condition, the MLE of (d', c) from counts is exactly what the Week 2 formulas give *without* correction, because the model is saturated (2 parameters, 2 rates) and the MLE reproduces the observed rates. The log-linear correction is *not* MLE; it is closer to a Bayesian estimate with a weak prior (Week 9).

For an engineer

- This is maximum-likelihood estimation exactly as in estimation theory (ML channel or frequency estimation, ML sequence detection). The only change is the noise model: binomial counts instead of AWGN.
- LS = ML under AWGN (Section 4) is a result you already use: with white Gaussian noise of known, equal variance, the ML estimator minimizes the squared error.
- The Fisher information and the asymptotic variance of the MLE (Advanced box) are the same quantities as in the Cramér–Rao lower bound. Week 5 uses them to compute standard errors.
- Comparing a constrained and a free model with $2(\ell_{\text{free}} - \ell_{\text{constrained}})$ is the generalized likelihood-ratio test (GLRT): both hypotheses have unknown parameters that are replaced by their ML estimates.
- Payoffs and criterion shifts: in Bayesian detection the optimal threshold moves with the costs and priors. Section 6 tests whether *people* move only their threshold (c) when the payoffs

change, while their "SNR" (d') stays the same.

For the instructor

- The one sentence to repeat: "the likelihood is the probability of the data we observed, as a function of the parameters". Write it on the board and leave it there all session.
- Plot it. Draw $L(\theta)$ for 7 of 10 (worked example, Part 1) as a curve over θ from 0 to 1. Ask: where is the peak? Does the area under the curve equal 1? (No.) Then draw it for 70 of 100: same peak, much narrower curve. That narrowness is precision, the topic of Week 5.
- Use your background. Many students find MLE abstract; you can present it as "the estimation method engineers use when the noise is not Gaussian".
- Typical questions.
  - *"Is the likelihood the probability that θ is correct?"* No. That needs a prior (Week 9).
  - *"Why minimize NLL instead of maximizing L?"* Same answer, but sums of logs avoid numerical underflow, and optimizers are written to minimize.
  - *"Is a non-significant G^2 proof the model is right?"* No: it only means the misfit is not larger than the noise, and with few data, real misfit can hide.
- Time management. Section 4 (Gaussian case) can be shown quickly as "LS is a special case". Spend the saved time on Section 5 (G^2).

1.1.5 Worked example

Part 1: likelihood of a single proportion

$y = 7$ recalled out of $n = 10$:

θ	0.5	0.6	0.7	0.8
$L(\theta) = \binom{10}{7}\theta^7(1-\theta)^3$	0.117	0.215	0.267	0.201
$\ell(\theta)$	−2.144	−1.537	−1.321	−1.603

The maximum is at $\hat{\theta} = 7/10 = 0.7$. Note that the likelihood values do not sum to 1 across θ: likelihood is not a probability distribution over parameters.

Part 2: a two-interval forgetting experiment

$y = (30, 13)$ items recalled out of $n = 50$ at $t = (1, 7)$ days. Model: $f(t) = Ae^{-\lambda t}$.

Step 1: NLL for a candidate. For $(A, \lambda) = (0.70, 0.15)$: $f(1) = 0.7e^{-0.15} = 0.6025$, $f(7) = 0.7e^{-1.05} = 0.2450$.

$$\ell = \ln \text{Binom}(30; 50, 0.6025) + \ln \text{Binom}(13; 50, 0.2450) = -2.167 + (-2.088) = -4.255,$$

so NLL $= 4.255$.

Step 2: compare. For $(0.70, 0.12)$: $f = (0.621, 0.302)$ and NLL $= 4.488$, which is worse. The first candidate makes the data more probable.

Step 3: the MLE. With 2 parameters and 2 data points the model can match both observed proportions exactly: $Ae^{-\lambda} = 0.60$ and $Ae^{-7\lambda} = 0.26$. Dividing gives $e^{6\lambda} = 0.60/0.26$, so $\hat{\lambda} = \ln(2.308)/6 = 0.139$ and $\hat{A} = 0.60\, e^{0.139} = 0.690$, with NLL $= 4.225$ (the minimum). A numerical optimizer gives the same values.

Interpretation. The MLE says this participant recalls about 69% at the start and loses about 13% of what remains per day ($1 - e^{-0.139}$). Because $k = N$ here, $G^2 = 0$: the model *cannot* misfit, so the perfect fit carries no evidence for the exponential form. Only data with more cells than parameters can test a model.

1.1.6 Lab (2 hours)

Goals

1. Plot a likelihood function and see why we work with log-likelihoods.
2. Fit exponential and power forgetting models by MLE; compare with least squares; evaluate with G^2.
3. Verify numerically that Gaussian MLE equals least squares.
4. Fit a constrained multi-condition SDT model and interpret what it says about payoffs.

Timing: Part A 15 min · Part B 35 min · Part C 15 min · Part D 35 min · Part E 20 min (exercises).

Setup

Code · کد: `code/week04.py`, cell 1 · سلول 1

Part A: the likelihood function (15 min)

Code · کد: `code/week04.py`, cell 2 · سلول 2

Expected output:

```
Area under L(theta): 0.091 (not 1: L is not a distribution over theta)
Product of 2000 trial probabilities: 0.0
Sum of their logs: -1239.52
```

Part B: MLE for forgetting curves (35 min)

> Code · کد: `code/week04.py`, cell 3 · سلول 3

Expected output:

```
model method      A   rate     NLL    SSE
  exp    MLE 0.6377 0.0458 29.8123 0.1100
  exp     LS 0.6568 0.0566 30.6983 0.1061
  pow    MLE 0.8773 0.4214 19.8303 0.0262
  pow     LS 0.8712 0.4251 19.8585 0.0259
True (power): A = 0.9, d = 0.5
```

What to notice: - Each method wins on its *own* criterion: MLE has the lowest NLL, LS the lowest SSE. - For the (true) power model the two methods agree closely. - For the exponential model the two methods differ more (decay rate 0.046 vs. 0.057). The binomial likelihood gives more weight to the precisely measured points near the floor (long delays), so the ML curve decays more slowly in order to stay closer to them.

> Warning: a real optimizer trap. Replace `"L-BFGS-B"` with `"Nelder-Mead"` in the ML fit above (keeping `bounds=bounds`) and rerun. For the exponential model, scipy's bounded Nelder–Mead collapses against the bound `A=1` and reports `success` with NLL = 79.5, far worse than the true minimum of 29.8 (it is even worse than the LS estimate's NLL). Bounded simplex methods can degenerate when vertices are pushed onto a bound. Two quick checks would catch this: an estimate sitting exactly on a bound, and a "MLE" whose NLL is *higher* than that of some other parameter values you have tried. Multi-start (Week 3) is the general cure.

Now evaluate absolute fit with the deviance:

> Code · کد: `code/week04.py`, cell 4 · سلول 4

Expected output:

```
exp: G2 =  26.27, df = 6, p = 0.0002
pow: G2 =   6.31, df = 6, p = 0.3895
```

Interpret: the exponential model misfits far more than binomial noise allows ($p = .0002$): we can reject it as a description of these data. The power model's misfit is consistent with sampling noise. That does not prove it is the right model; it means these data give no reason to reject it.

Code · کد: `code/week04.py`, cell 5 · سلول 5

Part C: Gaussian MLE equals least squares (15 min)

Treat the proportions as continuous and assume Gaussian noise with a free σ (a third parameter). The estimates of A and d should match least squares, and $\hat{\sigma}$ should equal $\sqrt{\text{SSE}/N}$.

Code · کد: `code/week04.py`, cell 6 · سلول 6

Expected output:

```
Gaussian MLE: A = 0.8712, d = 0.4251, sigma = 0.0569
Least squares: A = 0.8711, d = 0.4251, sqrt(SSE/N) = 0.0569
```

Part D: a constrained multi-condition SDT model (35 min)

The experiment. Participants detect a faint tone under three payoff schemes: liberal (hits rewarded), neutral, and conservative (false alarms penalized). 80 signal and 80 noise trials per condition. We simulate one participant whose sensitivity is the same in all conditions and whose criterion shifts with the payoffs.

Code · کد: `code/week04.py`, cell 7 · سلول 7

Expected output:

```
      condition  hits  n_signal  fas  n_noise
0       liberal    73        80   33       80
1       neutral    58        80   16       80
2  conservative    44        80    7       80

Shared d':   d' = 1.496, c = [-0.555  0.124  0.618], NLL = 13.134
Separate d': d' = [1.577 1.439 1.482], c = [-0.568  0.122  0.615], NLL =
 ↪  13.043

Likelihood-ratio statistic = 0.18, df = 2, p = 0.913
```

Note that the estimated `d'` (about 1.5) is higher than the generating value of 1.2. With 80 trials per cell that is ordinary sampling error. Week 5 shows how to put a confidence interval around it.

Interpret (write in your notebook): 1. Does constraining d' to be equal across payoff conditions cost much likelihood? What do you conclude about the effect of payoffs? 2. The separate-d' estimates differ from each other by a few tenths. Why is that not evidence that payoffs change sensitivity? 3. Check that the separate-d' model reproduces the observed hit and false-alarm rates exactly. Why must it?

Exercises (increasing difficulty)

Exercise 1 $\star$. Derive by hand the MLE of θ for a Bernoulli sequence with y successes in n trials, and check it numerically with `minimize_scalar` for $y = 13$, $n = 40$.

Exercise 2 $\star$. In Part B, refit both models with the Bernoulli likelihood on item-level data (expand each count into 0/1 responses). Show that the estimates are identical to the binomial fits and that the NLLs differ by a constant. What is the constant?

Exercise 3 $\star\star$ (zero counts). Set the last two counts to 0 (`recalled[-2:]=0`). Which fitting methods from Weeks 2–4 still work? Fit both models by MLE and comment on what happens to the estimates and to G^2.

Exercise 4 $\star\star$ (a third SDT model). Fit a model in which the criterion is the *same* in all conditions but d' differs ($k = 4$). Compare its NLL with the shared-d' model. Both have $k = 4$; why can you compare them directly by NLL here? Which one is psychologically more plausible for a payoff manipulation?

Exercise 5 $\star\star\star$ (Advanced: LS vs. MLE by simulation). Simulate 500 datasets from the power model ($A = 0.9$, $d = 0.5$, the 8 intervals, $n = 50$), fit each by LS and by MLE, and compare the two estimators of d: mean (bias), SD, and root-mean-squared error. Which is better here? Repeat with $n = 10$. (This is good preparation for A2.)

Lab check-off

Show: (1) the LS-vs-MLE table and G^2 results from Part B, (2) the Gaussian-vs-LS comparison from Part C, and (3) your answers to the interpretation questions of Part D.

1.1.7 Connection to AI & LLMs

LLMs are trained by maximum likelihood. The "cross-entropy loss" used to train every language model is the negative log-likelihood of the next token, averaged over a huge text corpus. Training an LLM is therefore an MLE problem with billions of parameters, solved by gradient descent instead of Nelder–Mead. Two quantities you meet in AI papers are this week's quantities in disguise: loss = NLL per token, and perplexity = $e^{\text{NLL per token}}$, the effective number of equally likely choices the model is "confused" between.

The simplest language model, a bigram model, has a closed-form MLE: $P(w_2 \mid w_1) = \text{count}(w_1 w_2)/\text{count}(w_1)$. It shows a familiar problem. Any word pair never seen in training gets probability 0, so a single new pair makes the likelihood of a whole sentence 0 (NLL = ∞). Smoothing solves this, in the same way that the log-linear correction of Week 2 solved hit rates of 1.

Code · کد: `code/week04.py`, cell 8 · سلول 8

Expected output:

```
P(cat | the) = 0.333
'the dog sat on the mat .'   MLE      NLL =  3.584  perplexity =   1.82
'the dog sat on the mat .'   add-one  NLL =  9.537  perplexity =   4.90
'the dog saw the cat .'      MLE      NLL =    inf  perplexity =    inf
'the dog saw the cat .'      add-one  NLL =  9.624  perplexity =   6.85
```

Try it (no API needed). - ⋆ Which word pair in "the dog saw the cat ." was never seen in the corpus? Check by hand that the MLE of $P(\text{cat} \mid \text{the})$ is 1/3. - ⋆⋆ Smoothing makes the *training* sentence less likely (NLL 3.6 → 9.5) but makes the new sentence possible. Replace add-one with add-k and choose k by leave-one-sentence-out cross-validation (a preview of Week 6). - ⋆⋆ Discussion: the MLE of a bigram model reproduces the training corpus as closely as possible. Why is "maximizing likelihood on internet text" not the same as "learning to tell the truth"?

1.1.8 Common mistakes & misconceptions

1. "The likelihood is the probability that the parameters are true." It is the probability (density) of the *data* given the parameters. It does not integrate to 1 over $\boldsymbol{\theta}$.
2. Multiplying probabilities instead of summing logs. The product underflows to 0 for realistic numbers of trials (Part A).
3. Maximizing when the optimizer minimizes. `minimize` needs the negative log-likelihood. Forgetting the minus sign sends the optimizer to the *worst* parameters.
4. Not clipping probabilities. Predicted probabilities of exactly 0 or 1 produce $-\infty$ log-likelihoods and crash the optimizer.

5. Comparing NLLs across different datasets or data codings. NLL depends on the number of observations and on constants (e.g., binomial coefficients). Compare only models fitted to the same data, coded the same way.
6. Using least squares for choices or counts by default. It ignores the data's noise structure. Use the likelihood that matches how the data were generated.
7. Reading a non-significant G^2 as proof the model is correct. It only says the misfit is not detectably larger than sampling noise at this sample size.
8. Forgetting that a saturated or exactly-determined model always fits perfectly (worked example, Part D separate-d' model). A perfect fit is then uninformative.

1.1.9 Check your understanding

1. In one sentence each: what varies and what is fixed in a probability distribution vs. a likelihood function? Answer Probability: parameters fixed, data vary, and it sums/integrates to 1 over data. Likelihood: data fixed at the observed values, parameters vary, and it need not integrate to 1 over parameters.

2. Give two reasons for working with log-likelihoods. Answer (i) Numerical: products of many probabilities underflow, while sums of logs do not. (ii) The log is monotone, so the maximizer is unchanged, and sums are easier to differentiate and optimize.

3. Under what assumptions is least squares equivalent to maximum likelihood? Show the key step. Answer Independent Gaussian errors with equal variance σ^2. Then $\ell = -\frac{N}{2}\ln(2\pi\sigma^2) - \text{SSE}/(2\sigma^2)$, so for any σ, maximizing ℓ over $\boldsymbol{\theta}$ is minimizing SSE.

4. A model with 2 parameters is fitted to 8 binomial cells and gives $G^2 = 4.1$. What are the df, and what do you conclude? Answer df $= 8 - 2 = 6$. A χ^2_6 has mean 6, so 4.1 is well within the range expected from sampling noise ($p \approx 0.66$). There is no evidence of misfit, but that does not prove the model correct.

5. Why does MLE with a binomial likelihood weight data points differently than plain least squares does? Answer The binomial log-likelihood reflects the variance $f(1-f)/n$: points predicted near 0 or 1 (small variance) are fitted more tightly, and points with larger n count more. Plain least squares weights every squared error equally.

6. In the payoff experiment, what psychological hypothesis does the shared-d' model express, and how would you test it? Answer That payoffs change only response bias (criterion), not perceptual sensitivity. Test it by comparing the maximized log-likelihood with the separate-d' model: $2(\ell_6 - \ell_4) \sim \chi^2_2$ approximately (likelihood-ratio test), or with AIC/BIC (Week 7).

7. You fit a model and the optimizer returns NLL $= -35.2$ (negative). Is something wrong? Answer Not necessarily. For continuous data, densities can exceed 1 (e.g., a narrow Gaussian), so log-densities can be positive and the NLL negative. For discrete data (binomial), probabilities are ≤ 1, so the NLL must be ≥ 0, and a negative value would indicate a bug.

1.1.10 Assignment

A2: Least squares vs. maximum likelihood (1.0 point of 20; released Week 4, due at the start of the Week 6 lecture)

Data. Generate the dataset with exactly this code (everyone gets the same data):

> Code · کد: `code/week04.py`, cell 9 · سلول 9

Tasks.

1. Fit (2 pts). Fit the exponential and power models to these data by (a) least squares on proportions and (b) binomial MLE. Report all estimates in one table with SSE and NLL.
2. Evaluate (2 pts). Compute G^2, df and p for each model's MLE fit. Plot the data with 95% binomial intervals, both fitted curves, and residuals. Which model, if any, can be rejected?
3. Simulation study (3 pts). Treat the power model with $A = 0.95$, $d = 0.7$ as the truth. Simulate 500 datasets with the same design, fit each with LS and MLE, and report the bias, SD and RMSE of $\hat{d}$ for each method. Show histograms. Repeat with $n = 100$ items and describe what changes.
4. Interpret (2 pts, max. 300 words). Which estimator would you recommend for recall data, and why? Relate your answer to the noise structure of binomial data and to the near-floor counts (1–2 of 20) at the longest delays. What would happen to the Week 2 linearization method if one of those counts were 0?
5. Code quality (1 pt). Reproducible, commented notebook; seeds fixed; functions reused rather than copy-pasted.

Late submissions lose 10% per day (max. 3 days).

1.1.11 Further reading

- Farrell & Lewandowsky (2018), chapter 4. Maximum likelihood parameter estimation: likelihood functions for common data types, log-likelihood, and fitting in practice. *(Main reading.)*
- Azimifar & Ahmadi (1396), *Shenasayi-ye amari-ye olgu*, vol. 1 (in Persian): the section on maximum likelihood parameter estimation, with the statistical-learning view [verify chapter].
- Lee & Wagenmakers (2014), chapter 3. Inferences with binomials. Read it now for the binomial set-up; we revisit it from the Bayesian side in Week 9.

Preview for Week 5: an estimate without an uncertainty is half a result. We compute standard errors and confidence intervals from the curvature of the likelihood and by bootstrapping, and we discover how averaging participants can create a "law" that no individual follows.